# FINS3616 International Business Finance Term

# Term 2, 2026

*Week 4 Reading Guide and Tutorial Questions*

Chs 7 and 8: Currency Futures and Options and Interest Rate and Currency Swaps

**Week 4 Readings (Shapiro)**

Chapter 7 – *Currency Futures and Options Markets*

7.1 Futures Contracts

7.2 Currency Options

7.3 Reading Currency Futures and Options Prices

Chapter 8 –Interest Rate and Currency Swaps

*What is a currency swap?*

## Tutorial Questions

*Chapter 7:*

Conceptual Questions 2, 3, 4, 5

Futures: Problems 1, 2, 3

Options: Problems 5, 6

*Chapter 8*

Questions 2 and 3

Chapter 8, Problem 3

***Chapter 7***

**Chapter 7, Question 2**

*2. What are the basic differences between forward and futures contracts? Between futures and options contracts?*

**Chapter 7, Question 3**

*3. A forward market already existed, so why was it necessary to establish currency futures and currency options contracts?*

**Chapter 7, Question 4**

*4. Suppose that Texas Instruments must pay a French supplier €10 million in 90 days.*

*a. Explain how TI can use currency futures to hedge its exchange risk. How many futures contracts will TI need to fully protect itself?*

*b. Explain how TI can use currency options to hedge its exchange risk. How many options contracts will TI need to fully protect itself?*

*c. Discuss the advantages and disadvantages of using currency futures versus currency options to hedge TI's exchange risk.*

**Chapter 7, Question 5**

*5. Suppose that Bechtel Group wants to hedge a bid on a Japanese construction project. But because the yen exposure is contingent on acceptance of its bid, Bechtel decides to buy a put option for the ¥15 billion bid amount rather than sell it forward. In order to reduce its hedging cost, however, Bechtel simultaneously sells a call option for ¥15 billion with the same strike price. Bechtel reasons that it wants to protect its downside risk on the contract and is willing to sacrifice the upside potential in order to collect the call premium. Comment on Bechtel's hedging strategy.*

**Chapter 7, Problem 1**

*1. On Monday morning, an investor takes a long position in a pound futures contract that matures on Wednesday afternoon. The agreed upon price is $1.78/£ for £62,500. At the close of trading on Monday, the futures price has risen to $1.79/£. At Tuesday close, the price rises further to $1.80/£. At Wednesday close, the price falls to $1.785/£, and the contract matures. The investor takes delivery of the pounds at the prevailing price of $1.785/£. Detail the daily settlement process (The Table is taken from Chapter 7 of the Textbook). What will be the investor's profit (loss)?*

**ANSWER**

| Time | Action | Cash Flow |
|---|---|---|
| Monday morning | Investor buys pound futures contract that matures in two days. Price is $1.78. | None |
| Monday close | Futures price rises to $1.79. Contract is marked-to-market. | Investor receives 62,500 x (1.79 - 1.78) = $625. |
| Tuesday close | Futures price rises to $1.80. Contract is marked-to-market. | Investor receives 62,500 x (1.80 - 1.79) = $625. |
| Wednesday close | Futures price falls to $1.785. (1) Contract is marked-to-market. (2) Investor takes delivery of £62,500. | (1) Investor pays 62,500 x (1.80 -1.785) = $937.50 (2) Investor pays 62,500 x 1.785 = $111,562.50. |

**Net profit is $1,250 - 937.50 = $312.50.**

**Chapter 7, Problem 2**

*2. Suppose that the forward ask price for March 20 on euros is $0.9127 at the same time that the price of IMM euro futures for delivery on March 20 is $0.9145. How could an arbitrageur profit from this situation? What will be the arbitrageur's profit per futures contract (size of contract is €125,000)?*

**Chapter 7, Problem 3**

*3. Suppose that Dell buys a Swiss franc futures contract (contract size is SFr 125,000) at a price of $0.83. If the spot rate for the Swiss franc at the date of settlement is SFr 1 = $0.8250, what is Dell's gain or loss on this contract?*

**Chapter 7, Problem 5**

*5. Citigroup sells a call option on euros (contract size is €500,000) at a premium of $0.04 per euro. If the exercise price is $0.91/€ and the spot price of the euro at the date of expiration is $0.93/€, what is*

**Chapter 7, Problem 6**

*6. Suppose you buy three June PHLX call options with a 90 strike price at a price of 2.3 (¢/€). (Contract size is €62,500).*

- *a. What would be your total dollar cost for these calls, ignoring broker fees?*
- *b. After holding these calls for 60 days, you sell them for 3.8 (¢/€). What is your net profit on the contracts assuming that brokerage fees on both entry and exit were $5 per contract and that your opportunity cost was 8% per annum on the money tied up in the premium?*

***<u>Chapter 8</u>***

## <u>Chapter 8, Question 2</u>

*2. What is a currency swap?*

## <u>Chapter 8, Question 3</u>

*3. Comment on the following statement. "In order for one party to a swap to benefit, the other party must lose."*

***<u>Chapter 8, Problem 3</u>***

*2. In May 1988, Walt Disney Productions sold to Japanese investors a 20 year stream of projected yen royalties from Tokyo Disneyland. The present value of that stream of royalties, discounted at 6 percent (the return required by the Japanese investors), was ¥93 billion. Disney took the yen proceeds from the sale, converted them to dollars, and invested the dollars in bonds yielding 10 percent. According to Disney's chief financial officer, Gary Wilson, "In effect, we got money at a 6 percent discount rate, reinvested it at 10 percent, and hedged our royalty stream against yen fluctuations all in one transaction."*

- *a. At the time of the sale, the exchange rate was ¥124 = $1. What dollar amount did Disney realize from the sale of its yen proceeds?*
- *b. Demonstrate the equivalence between Walt Disney's transaction and a currency swap.*
- *c. Comment on Gary Wilson's statement. Did Disney achieve the equivalent of a free lunch through its transaction?*